

MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Sets, Numbers & Intervals

(English Medium)

EXERCISE (O-1)

Straight Objective Type

1. Which of the following statements is true ?
 (A) $3 \subseteq \{1, 3, 5\}$ (B) $3 \in \{1, 3, 5\}$ (C) $\{3\} \in \{1, 3, 5\}$ (D) $\{3, 5\} \in \{1, 3, 5\}$
FM0005
2. Which of the following is a null set ?
 (A) $A = \{x : x > 1 \text{ and } x < 1\}$ (B) $B = \{x : x + 3 = 3\}$
 (C) $C = \{\phi\}$ (D) $D = \{x : x \geq 1 \text{ and } x \leq 1\}$
FM0006
3. If A and B be any two sets, then $(A \cap B)'$ is equal to-
 (A) $A' \cap B'$ (B) $A' \cup B'$ (C) $A \cap B$ (D) $A \cup B$
FM0001
4. Let A and B be two sets in the universal set. Then $A - B$ equals-
 (A) $A \cap B'$ (B) $A' \cap B$ (C) $A \cap B$ (D) none of these
FM0002
5. If $A \subseteq B$, then $A \cap B$ is equal to-
 (A) A (B) B (C) A' (D) B'
FM0003
6. If A and B are any two sets, then $A \cup (A \cap B)$ is equal to-
 (A) A (B) B (C) A' (D) B'
FM0004
7. If A and B are not disjoint, then $n(A \cup B)$ is equal to-
 (A) $n(A) + n(B)$ (B) $n(A) + n(B) - n(A \cap B)$
 (C) $n(A) + n(B) + n(A \cap B)$ (D) $n(A) \cdot n(B)$
FM0007
8. In a college of 300 students every student reads 5 newspapers and every newspaper is read by 60 students. The number of newspapers is
 (A) at least 30 (B) at most 20 (C) exactly 25 (D) none of these
FM0031
9. Number of integers in between 200 to 2023 (both 200 and 2023 including) are divisible by 2 or 5
 (A) 1093 (B) 1094 (C) 1275 (D) 1277
ST0047
10. Number of integers in between 500 to 2023 are divisible by 3 or 5 but not multiple of 2
 (A) 356 (B) 355 (C) 710 (D) 711
ST0048
11. Let $S = \{1, 2\}$, number of elements in the set $\{(A, B) : A \cup B = S\}$
 (A) 4 (B) 6 (C) 9 (D) 10
ST0049
12. How many integers between 100 and 999 (all three digits of integer are not distinct) inclusive, have the property that some permutation of its digits is a multiple of 11 between 100 and 999? For example both 121 and 211 have this property
 (A) 48 (B) 44 (C) 42 (D) 40
ST0050

13. How many of the following statements are correct

P : In roster form, the order in which the elements are listed is immaterial

Q : While writing a set in roster form, an element is not generally repeated.

R : The collection of five most renowned mathematicians of the world is not a set.

S : A collection of most dangerous animals of the world is a set.

- (A) 1 (B) 2 (C) 3 (D) 4

ST0051

14. How many of the following statements are sets.

(i) The collection of all the months of a year beginning with the letter J.

(ii) The collection of ten most talented writers of india.

(iii) A team of eleven best-cricket batsmen of the world.

(iv) The collection of all boys in your class.

(v) The collection of all natural numbers less than 100.

- (A) 1 (B) 2 (C) 3 (D) 4

ST0052

15. How many of the following statements are correct?

(1) Let $A = \{x : 1 < x < 2, x \text{ is a natural number}\}$. Then A is the empty set, because there is no natural number between 1 and 2.

(2) $B = \{x : x^2 - 2 = 0 \text{ and } x \text{ is rational number}\}$. Then B is the empty set because the equation $x^2 - 2 = 0$ is not satisfied by any rational value of x.

(3) $C = \{x : x \text{ is an even prime number greater than } 2\}$. Then C is the empty set, because 2 is the only even prime number

(4) $D = \{x : x^2 = 4, x \text{ is odd}\}$. Then D is the empty set, because the equation $x^2 = 4$ is not satisfied by any odd value of x .

- (A) 1 (B) 2 (C) 3 (D) 4

ST0053

16. How many of the following statements are correct ?

(1) Let W be the set of the days of the week. Then W is finite.

(2) Let S be the set of solutions of the equation $x^2 - 16 = 0$. Then S is finite.

(3) Let G be the set of points on a line. Then G is infinite.

- (A) 0 (B) 1 (C) 2 (D) 3

ST0054

17. How many of the following statements are incorrect ?

(1) Let $A = \{1,2,3,4\}$ and $B = \{3,1,4,2\}$. Then $A = B$.

(2) Let A be the set of prime numbers less than 6 and P the set of prime factors of 30. Then A and P are equal, since 2, 3 and 5 are the only prime factors of 30 and also these are less than 6.

(3) A set does not change if one or more elements of the set are repeated. For example, the sets $A = \{1, 2, 3\}$ and $B = \{2, 2, 1, 3, 3\}$ are equal, since each element of A is in B and vice-versa. That is why we generally do not repeat any element in describing a set.

- (A) 0 (B) 1 (C) 2 (D) 3

ST0055

18. How many of the following statements are correct.

P : All infinite sets cannot be described in the roster form.

Q : Let W be the set of the days of the week. Then W is finite

R : Let $A = \{x : 1 < x < 2, x \text{ is a natural number}\}$. Then A is the empty set,

S : Let G be the set of points on a line. Then G is infinite

(A) 4 (B) 1 (C) 2 (D) 3

ST0056

19. How many of the following statements are correct ?

P. $A \subset B$ if $a \in A \Rightarrow a \in B$

Q. A is a subset of B if a is an element of A implies that a is also an element of B

R. ϕ is a subset of every set

S. Every set A is a subset of itself

(A) 4 (B) 1 (C) 2 (D) 3

ST0057

20. If $x + y = 1$ and $x^2 + y^2 = 2$ then the value of $x^4 + y^4$ equals

(A) 7 (B) 6 (C) $\frac{7}{2}$ (D) $\frac{19}{4}$

FM0032

21. If $x = 2^{1/3} - 2^{-1/3}$ then the value of $2x^3 + 6x$ is equal to

(A) 1 (B) 2 (C) 3 (D) 4

FM0033

22. Consider the operation "minus the reciprocal of, defined by " $a \diamond b = a - \frac{1}{b}$ ".

What is $((1 \diamond 2) \diamond 3) - (1 \diamond (2 \diamond 3))$?

(A) $-\frac{7}{30}$ (B) $-\frac{1}{6}$ (C) 0 (D) $\frac{1}{6}$

ST0058

23. Pranshi has written down one integer two times and another integer three times. The sum of the five numbers is 100, and one of the numbers is 28. The other number is

(A) 6 (B) 8 (C) 10 (D) 12

ST0059

24. If $x, y \in \mathbb{N}$ and $xy - 3x - 2y = 54$, then number of ordered pairs (x, y) is

(A) 7 (B) 8 (C) 12 (D) 24

ST0060

25. How many ordered triples (x, y, z) of positive integers satisfy $\text{lcm}(x, y) = 72$, $\text{lcm}(x, z) = 600$ and $\text{lcm}(y, z) = 900$?

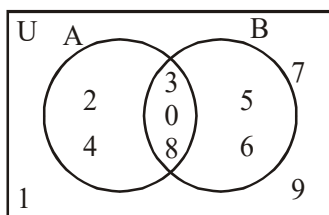
(A) 14 (B) 15 (C) 16 (D) 20

ST0061

EXERCISE (O-2)

Multiple Correct Answer Type

- If $U = \{1,2,3,4,5,6,7,8\}$, $A = \{1,2,3,5,6\}$ and $B = \{2,3,4,7,8\}$ which of the following are correct
 (A) $A \cup B = U$ (B) $B - A = \{1,5,6\}$
 (C) $A' \cup B = \{2,3,4,7,8\}$ (D) $(A - B)' = \{2,3,4,7,8\}$
ST0062
- If $A = \{1,2,3,4\}$, $B = \{3,4,5,6\}$, $C = \{5,6,7,8\}$ and $D = \{7,8,9,10\}$. Which of the following option(s) is/are true ?
 (A) $A \cup C = \{1,2,3,4,5,6,7,8\}$ (B) $B \cup D = \{3,4,5,6,7,8,9,10\}$
 (C) $A \cup B \cup D = \{1,2,3,4,5,6,7,8,9,10\}$ (D) $B \cup C \cup D = \{3,4,5,6,7,8,9,10\}$
ST0063
- Which of the following are correct?
 (A) $[2, 10] - \{2, 10\} = (2, 10)$ (B) $[3, 5] - \{5\} = (3, 5]$
 (C) $(-3, 5] - \{-3\} = (3, 5]$ (D) $[-2, 15] - (2, 15] = [-2, 2]$
ST0064
- If $A \subseteq B$ then which of the following option(s) is/are correct ?
 (A) $A' \subseteq B'$ (B) $B' \subseteq A'$ (C) $A \cap B' = \phi$ (D) $A' \cap B = \phi$
FM0036
- Which of the following intervals are subsets of $[2, 3]$?
 (A) $((0, 2.3] \cup (2.3, 3]) \cap (0, 2.3)$
 (B) $((2, 2.5] \cup (2.5, 4]) \setminus (0, 2.3)$
 (C) $((2, 2.5] \cup (2.5, 4]) \cap (0, 3)$
 (D) $((0, 2.3] \cup (2.2, 3]) \setminus (0, 2.3)$
ST0065
- Which of the following are correct ?
 (A) $[a, b]$ is an open interval from a to b , including a but excluding b .
 (B) $(a, b]$ is an open interval from a to b , including b but excluding a .
 (C) $[a, b)$ is an open interval from a to b , including b but excluding a .
 (D) (a, b) is an open interval from a to b , including a but excluding b .
ST0066
- From the adjoining Venn-diagram, find which of the following are correct :



- (A) $A' = \{1,5,6,7,9\}$ (B) $B' = \{1,2,4,7,9\}$
 (C) $(A \cap B)' = \{1,2,4,5,6,7,9\}$ (D) $(A - B)' = \{1,2,4,5,6,7,9\}$

ST0067

8. Which of the following option(s) is/are true ?

- (A) $A \cup A' = U$ (B) $A \subset B \Rightarrow A \cup B = B$
 (C) $A \subset B \Rightarrow A \cap B = A$ (D) $A \cap \phi = \phi$

ST0068

9. Which of the following option(s) is/are true ?

- (A) If $A \subset B$ and $A \neq B$, then A is called a proper subset of B
 (B) If $A \subset B$, then B is called superset of A .
 (C) $A = \{1,2,3\}$ is NOT a proper subset of $B = \{1,2,3,4\}$
 (D) If a set A has only one element, we call it a singleton set.

ST0069

10. In which of the following options, $A \neq B$:

- (A) $A = \{x : x + 2 = 3\}$, $B = \{x : x \in \mathbb{N} \text{ and is less than } 2\}$
 (B) $A = \{x : x \in \mathbb{N} \text{ and } 3x - 1 < 2\}$, $B = \{x : x \in \mathbb{W} \text{ and } 3x - 1 < 2\}$
 (C) $A = \{x : x \in \mathbb{N} \text{ and is prime factor of } 36\}$, $B = \{1,2,3,4,6,9,12\}$
 (D) $A = \{x : x \in \mathbb{I} \text{ and } x^2 \leq 4\}$, $B = \{x : x \in \mathbb{R} \text{ and } x^2 - 3x + 2 = 0\}$

ST0070

11. Which of the following are examples of the null set

- (A) Set of odd natural numbers divisible by 2
 (B) Set of even prime numbers
 (C) $\{x : x \text{ is a natural numbers } x < 5 \text{ and } x > 7\}$
 (D) $\{y : y \text{ is a point common to any two parallel lines}\}$

ST0071

12. Consider the set ϕ , $A = \{1,3\}$, $B = \{1,5,9\}$, $C = \{1,3,5,7,9\}$. Which of the following statements are correct ?

- (A) $\phi \subset B$ as ϕ is a subset of every set.
 (B) $A \not\subset B$ as $3 \in A$ and $3 \notin B$
 (C) $A \subset C$ as $1,3 \in A$ also belongs to C
 (D) $B \subset C$ as each element of B is also an element of C .

ST0072

13. Which of the following options are correct ?

- (A) The set Q of rational numbers is a subset of the set R of real numbers, and we write $Q \subset R$.
 (B) If A is the set of all divisor of 56 and B the set of all prime divisors of 56, then B is a subset of A and we write $B \subset A$.
 (C) Let $A = \{1,3,5\}$ and $B = \{x : x \text{ is an odd natural number less than } 6\}$. Then $A \subset B$ and $B \subset A$ and hence $A = B$.
 (D) Let $A = \{a,e,i,o,u\}$ and $B = \{a,b,c,d\}$. Then A is not a subset of B , also B is not a subset of A .

ST0073

14. Which of the following options are incorrect
- (A) $\{2,3,4,5\}$ and $\{3,6\}$ are disjoint sets
- (B) $\{a,e,i,o,u\}$ and $\{a,b,c,d\}$ are disjoint sets
- (C) $\{2,6,10,14\}$ and $\{3,7,11,15\}$ are disjoint sets
- (D) $\{2,6,10\}$ and $\{3,7,11\}$ are disjoint sets

ST0074

15. Consider the following sets.

$$S_1 = \{x | x \in \mathbb{R}, 0 < x < 10\}$$

$$S_2 = \{x | x \in \mathbb{Z}, -10 \leq x \leq 10\}$$

$$S_3 = \{x | x \in \mathbb{Q}, 10 \leq x \leq 15\}$$

Choose the set of correct options.

- (A) $(S_2 \cap S_3) \cup (S_3 \cap S_1) = \{10\}$
- (B) $(S_2 \setminus S_1) \cap (S_3 \setminus S_1) = \phi$
- (C) The cardinality of the set $S_1 \cap S_2$ is 10
- (D) The cardinality of the set $(S_1 \cap S_2) \cup (S_2 \cap S_3)$ is 10

ST0075

16. Let $A, B, C \subseteq X$. Which of the following option(s) is/are true ?

- (A) $[A \setminus (A \cap B)] \cup [B \setminus (A \cap B)] \cup (A \cap B) = A \cup B$
- (B) $[(A \cup B)^c \cup B] \cap [B^c \cup (A \cup B)] = (A \setminus B)^c$
- (C) $[(A \cup B) \cap (A \cap C)] \cap B^c = (A \setminus B) \cap C^c$
- (D) $[A \cap (B^c \cap A^c)^c]^c \cup A = X$

ST0076

17. Let $A, B, C, D \subseteq X$. Which of the following option(s) is/are true ?

- (A) $(C \cup B)^c \setminus (A \cap (B \setminus C)) = B^c \cap C^c$
- (B) $A \cup [(A \cup (A \cup B)) \cap (B \cup (A \cap B))] = A \cup B$
- (C) $C^c \cup [D^c \cup (D \setminus C^c)^c] = C^c \cup D^c$
- (D) $[(A \cup B) \cup B^c] \cap [B^c \cap (A \cup B)] = A \setminus B$

ST0077

18. Let A be a set and let $B = \{(a, S) \in A \times P(A) : a \in S\}$

For which of the following option(s) B is not an empty set :

- (A) $A = \{1\}$ (B) $A = \{1, 2\}$ (C) $A = \{1, 2, 3\}$ (D) None of these

ST0078

19. If complex number $-3 + ix^2y$ and $x^2 + y + 4i$ are conjugate of each other. Then real value of x & y can be

- (A) $x = 1, y = -4$ (B) $x = -1, y = -4$ (C) $x = 1, y = 4$ (D) $x = -2, y = -3$

FM0038

20. If $a \bmod 3 = 1$ and $b \bmod 3 = 2$, where $a, b \in \mathbb{N}$, then choose the set of correct options.

($a \bmod b = c$, $a, b, c \in \mathbb{N}$; means $a = nb + c$ where $n \in \mathbb{N}$)

- (A) $(ab) \bmod 3 = 0$ (B) $(ab) \bmod 3 = 2$
- (C) $(a + b) \bmod 3 = 0$ (D) $(a + b) \bmod 2 = 1$

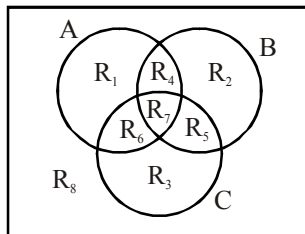
ST0079

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for 1 TO 3

R_i denotes region for $i = 1, 2, \dots, 8$ as shown in figure.



1. Which of the following option(s) is/are true ?

(A) $R_1 = A \cap (B \cup C)'$

(B) $R_2 = B \cap (A' \cap C')$

(C) $R_3 = (A \cup B)'$

(D) $R_8 = (A \cup B \cup C)'$

ST0081

2. Which of the following option(s) is/are true ?

(A) $R_4 = (A \cap B) \cap C'$

(B) $R_5 = (B \cap C) \setminus A$

(C) $R_6 = (A \cap C) \cap (A \cap B \cap C)'$

(D) $R_8 = A' \cap B' \cap C'$

ST0081

3. Which of the following option(s) is/are true ?

(A) $R_7 = A \cap B \cap C$

(B) $R_7 = (A' \cup B' \cup C')'$

(C) $R_3 = (A' \cap B') \cap C$

(D) $R_5 = (B \setminus C') \cap A'$

ST0082

Paragraph for 4 TO 6

Let S be the universal set, A, B & C are sets such that $|(A \cup B \cup C)'| = 1$ & $|S| = 8$.

4. If $|A \cap B \cap C|$ is maximum then which of the following option(s) is/are true ?

(A) $|A| = |B| = |C|$

(B) $A \cap B' = B \cap A' = A \cap C' = A' \cap C$

(C) $A \setminus B = (A \cup B \cup C) - (A \cap B \cap C)$

(D) $A' = S \setminus B$

ST0083

5. If $|A \cap B \cap C| = 6$

(A) If $A = B$, then $|C| = 7$

(B) If $A = B$ and $A \setminus C \neq \phi$, then $|C| = 6$

(C) If $A = B$ and $A \setminus C \neq \phi$, then $|A \cap B| = 7$

(D) If $A = B = C$ then $|A \cup B \cup C| = 7$

ST0084

6. If $|A| = |B| = |C|$, then which of the following option(s) is/are true ?

(A) Sum of minimum and maximum value of $|A \cap B \cap C|$ is 7.

(B) Let X be the set of all possible values of cardinality of $A \cap B \cap C$, then $|X| = 7$.

(C) Let X be the set of all possible values of cardinality of $A \cap B \cap C$, then $|X| = 6$.

(D) If $|A| = 4$, then $|A \cap B \cap C| \in \{0, 1, 2\}$.

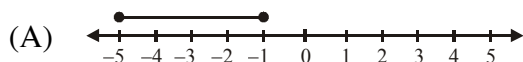
ST0085

Matrix Match Type

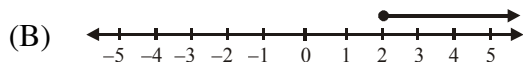
7.

Column-I

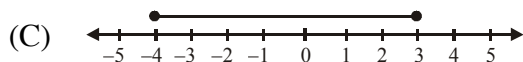
Column-II



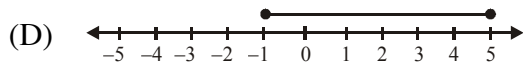
(P) $-1 \leq x \leq 5; x \in \mathbb{R}$



(Q) $-5 \leq x \leq -1; x \in \mathbb{R}$



(R) $x \geq 2; x \in \mathbb{R}$



(S) $-4 \leq x \leq 3; x \in \mathbb{R}$

ST0086

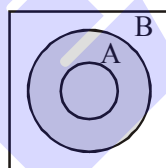
8. Given sets A and B; such that $A \subseteq B$. Match the Venn-diagrams in column-II with the sets I column-I.

Column-I

Column-II

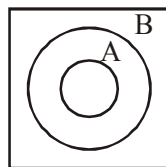
(A) $A \cup B$

(P)



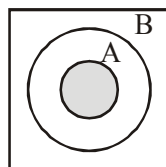
(B) $B' \cap A$

(Q)



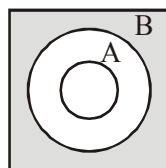
(C) $A \cap B$

(R)



(D) $(A \cup B)'$

(S)

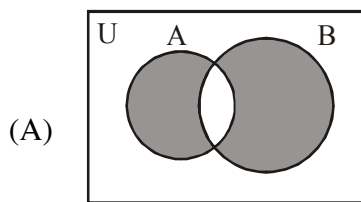


ST0087

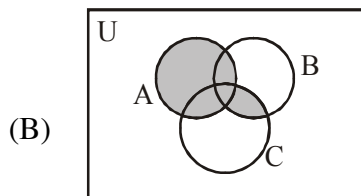
9.

Column-I

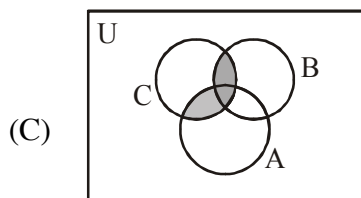
Column-II



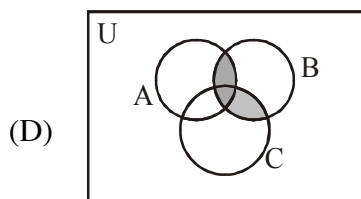
(P) $A \cup (B \cap C)$



(Q) $(A - B) \cup (B - A)$



(R) $(A \cap C) \cup (B \cap C)$



(S) $(A \cup C) \cap B$

ST0088

10. Match the following sets in set builder form in column-I with roster form in column-II

Column-I

Column-II

(A) $\left\{x : x \in \mathbb{I}, -\frac{1}{2} < x < \frac{9}{2}\right\}$

(P) $\{1, 2, 3, 4\}$

(B) $\{x : x \in \mathbb{N} \text{ and } 4x - 1 \leq 15\}$

(Q) $\left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots\right\}$

(C) $\left\{x : x = \frac{n}{n+1}, n \in \mathbb{N}\right\}$

(R) $\left\{\frac{1}{2}, \frac{3}{4}, \frac{5}{6}, \dots\right\}$

(D) $\left\{x : x = \frac{2n-1}{2n}, n \in \mathbb{N}\right\}$

(S) $\{0, 1, 2, 3, 4\}$

ST0089

EXERCISE (O-4)

Numerical Grid Type

1. An investigator interviewed 100 students to determine their preferences for the three drinks : milk (M), coffee (C) and tea (T). He reported the following : 10 students had all the three drinks M, C and T; 20 had M and C; 30 had C and T; 25 had M and T; 12 had M only; 5 had C only; and 8 had T only. Using a Venn diagram find how many did not take any of the three drinks.

FM0058

2. A survey shows that 63% of the Americans like cheese whereas 76% like apples. If $x\%$ of the Americans like both cheese and apples, number of integral values of x is.

FM0101

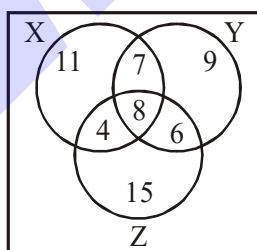
3. How many ordered pairs of integers which satisfy the equation $\frac{1}{m} + \frac{2}{n} = \frac{1}{2}$?

FM0102

4. Find the smallest natural number that leaves the remainders 3 and 4 when divided by 5 and 6 respectively.

ST0090

5. Suppose there are 100 people in a city C. Let X be the set of people who are engineers in the city C, Y be the set of people whose age is above 40 years in the city C, Z be the set of people who has an independent house in the city C. Now, study the following Venn diagram and answer the question given below.



Find the number of engineers who do not have an independent house ?

ST0091

6. Consider a set $S = \{a \mid a \in \mathbb{N}, a \leq 18\}$. Let R_1 and R_2 be sets of ordered pairs, defined as $R_1 = \{(x, y) \mid x, y \in S, y = 3x\}$ and $R_2 = \{(x, y) \mid x, y \in S, y = x^2\}$. Find the cardinality of the set $R_1 \setminus (R_1 \cap R_2)$.

ST0092

7. If $A = \{4, 5, 6\}$, $B = \{4, 7, 8\}$ and $C = \{4, 6, 7\}$, then find the cardinality of $(A - B) \times (B \cap C)$

ST0093

8. In DDPS – ALLEN, out of 400 students, 300 students opt for at least one subject out of Biology, Chemistry and Mathematics. If 120 students opted for Biology, 150 students opted for Chemistry, 200 students opted for Mathematics and 30 opted for all the three subjects, find the number of students who opted for exactly two subjects. ST0094
9. Let M be the set of positive integers less than or equal to 2000 such that the difference of any two numbers in M is neither 5 nor 8. Find the maximum number of elements in M . ST0095
10. In a Zoo, there are 6 Bengal white tigers and 6 Bengal royal tigers. Out of these tigers, 5 are males and 10 are either Bengal royal tigers or males. Find the number of female Bengal white tigers in the Zoo. ST0096

EXERCISE (JM)

1. If A , B and C are three sets such that $A \cap B = A \cap C$ and $A \cup B = A \cup C$, then :-
 (1) $B = C$ (2) $A \cap B = \phi$ (3) $A = B$ (4) $A = C$ FM0112
2. In a class of 140 students numbered 1 to 140, all even numbered students opted mathematics course, those whose number is divisible by 3 opted Physics course and those whose number is divisible by 5 opted Chemistry course. Then the number of students who did not opt for any of the three courses is :
 (1) 102 (2) 42 (3) 1 (4) 38 FM0113
3. Two newspapers A and B are published in a city. It is known that 25% of the city populations reads A and 20% reads B while 8% reads both A and B . Further, 30% of those who read A but not B look into advertisements and 40% of those who read B but not A also look into advertisements, while 50% of those who read both A and B look into advertisements. Then the percentage of the population who look into advertisement is :-
 (1) 12.8 (2) 13.5 (3) 13.9 (4) 13 FM0114
4. Let $\mathbb{Z}$ be the set of integers. If $A = \left\{ x \in \mathbb{Z} : 2^{(x+2)(x^2-5x+6)} = 1 \right\}$ and $B = \{ x \in \mathbb{Z} : -3 < 2x - 1 < 9 \}$, then the number of subsets of the set $A \times B$, is:
 (1) 2^{18} (2) 2^{10} (3) 2^{15} (4) 2^{12} FM0115

EXERCISE (JA)

1. If X and Y are two sets, then $X \cap (X \cup Y)^c$ equals

- (a) X (b) Y (c) ϕ (d) none of these

FM0116

2. If $A = \{1, 2, 3, 4, 5\}$, then the number of proper subsets of A is-

- (1) 120 (2) 30 (3) 31 (4) 32

FM0120

3. If A, B, C be three sets such that $A \cup B = A \cup C$ and $A \cap B = A \cap C$, then -

- (1) $A = B$ (2) $B = C$ (3) $A = C$ (4) $A = B = C$

FM0121

4. Sets A and B have 3 and 6 elements respectively. What can be the minimum number of elements in $A \cup B$?

- (1) 3 (2) 6 (3) 9 (4) 18

FM0122

ANSWER KEY

Do yourself-1

- | | | |
|-------------------------|--|----------------|
| 1. All are true | 2. C | 3. D |
| 4. (A-P, B-R, C-Q, D-S) | 5. (A-P, B-S, C-Q, D-R) | 6. 8 7. 2 |
| 8. 7 | 9. (A-iii-I ; B-i-II; C-ii-III, D-iv-IV) | |
| 10. (C) | 11. (3) | 12. 2 13. 3 |

Do yourself-2

- | | | |
|---------------------------------------|------|------|
| 2. (A-R, B-P, C-Q, D-S) | 3. 2 | 4. 4 |
| 5. (A)→(S); (B)→(P); (C)→(Q); (D)→(P) | 6. 2 | 7. 4 |

Do yourself-3

- | | | | | | | | |
|-------|-------|-------|-----------------------------|-------|-------|------|------|
| 2. C | 3. B | 4. B | 5. B | 6. C | 7. C | 8. D | 9. B |
| 10. D | 11. B | 12. B | 13. P-30, Q-19, R-60, S-No. | 14. 4 | 15. 2 | | |
| 16. 2 | 17. 8 | 18. B | 19. A, B, D | 20. A | | | |

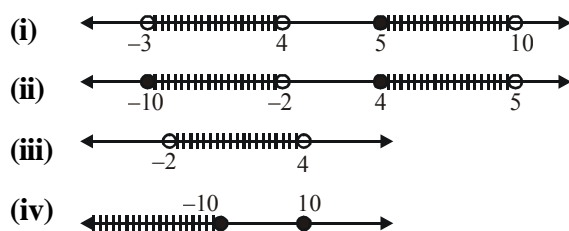
Do yourself-4

1. $a = 3, b = 4$
2. (i) $\frac{2}{11}$ (ii) $\frac{16}{99}$ (iii) $\frac{419}{990}$
4. (i) $x + y, x - y, xy, x/y$ are rational numbers.
 (ii) $x + y, x - y, xy, x/y$, all are may or may not be rational.
 (iii) $x + y, x - y, xy, x/y$ are irrational numbers.
6. x is irrational number (Non terminating and non-recurring)

Do yourself-5

1. $(x,y) \equiv (1,20); (2,10); (4,5); (5,4); (10,2); (20,1)$
2. (i) $(x,y) \equiv (2,2)$ (ii) $(x,y) \equiv (5,13); (6,4); (7,1)$
3. $(x,y) \equiv (8,56); (14,14); (56,8)$
4. 651 5. 293 6. 4 7. 30 8. 3

Do yourself-6



EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	A	B	A	A	A	B	C	B	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	C	C	D	D	A	A	A	C
Que.	21	22	23	24	25					
Ans.	C	A	B	C	B					

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C,D	A,B,C,D	A,D	B,C	C,D	A,B	A,B,C	A,B,C,D	A,B,D	B,C,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,C,D	A,B,C,D	A,B,C,D	A,B	A,D	A,B,C,D	A,B,C,D	A,B,C	A,B	B,C

EXERCISE # O-3

Que.	1	2	3	4	5	6				
Ans.	A,B,D	A,B,C,D	A,B,C,D	A,B,C,D	B,C,D	A,B,D				
Q.7	A	B	C	D	Q.8	A	B	C	D	
	Q	R	S	P		P	Q	R	S	
Q.9	A	B	C	D	Q.10	A	B	C	D	
	Q	P	R	S		S	P	Q	R	

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	20	25	7	28	18	5	4	110	770	2

EXERCISE # JEE-MAIN

Que.	1	2	3	4						
Ans.	1	4	3	3						

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4						
Ans.	C	3	2	2						